Capacity or Classifier? What Expansion Triggers Detect in Task-Free Continual Adapter Learning, and When Expansion Pays

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Abstract

Dynamic-expansion continual learners add an adapter when a trigger—a loss spike, a representation shift, a prospective-forgetting estimate, or the utility of a fresh module—signals that existing capacity is inadequate. We ask what these triggers detect and when expanding is worth it. At the arrival of labels that the current module's classifier has never trained on, the excess loss depends on the input only through the old labels' logits, so loss-based triggers and one-step fresh-module utility are *classifier-novelty* detectors. In a pre-registered study (ten text streams, three held out; DistilBERT, a held-out BERT backbone and ViT-B/16), every loss-spike and fresh-utility expansion in class-incremental streams fell at package-level label novelty. We then measure the value of expansion directly against two references, a single adapter that keeps learning and first-session adaptation. When the learner keeps exemplars from which its classifier statistics are refreshed, oracle expansion at the true boundaries costs 7–11 points in class-incremental streams relative to a single adapter, is neutral under domain shift and mixed under concept drift, although it beats first-session adaptation, the reference used in self-expansion ablations. When the learner is exemplar-free, so that a training adapter's prototypes go stale, the sign flips: [EXT2]. Refreshing stale prototypes without exemplars by semantic drift compensation [EXT4], and the same interaction appears inside the official implementation of a published self-expanding method (SEMA) [SEMA]. Expansion at classifier novelty is therefore a remedy for stale classifier statistics rather than for missing capacity, and its value depends on whether those statistics can be refreshed. We recommend that expansion triggers be evaluated by the measured value of expansion against both reference points, not by boundary precision.

1 Introduction

Parameter-efficient continual learning commonly grows a pool of adapters around a frozen pre-trained model (Zhou et al., 2024b; Wang et al., 2025; Wei et al., 2025). When task identities are unavailable—the task-free setting (Aljundi et al., 2019)—the learner must itself decide *when* to allocate a new module. Published decision rules monitor training-loss spikes or plateaus (Wei et al., 2025; Tathe, 2026), distribution-shift descriptors of intermediate representations (Wang et al., 2025), prospective forgetting after a virtual update (Julian et al., 2025), or the hindrance a new task places on forward learning (Feng et al., 2024). These signals are usually validated by *boundary precision*—whether expansions coincide with the hidden task boundaries of a benchmark—or by ablations against a model that adapts only on the first task.

Boundary precision and the *value* of expansion are different quantities. Most boundaries in standard benchmarks are class-incremental: new labels arrive with new inputs. A learner whose classifier can grow new output rows, on top of a pre-trained representation that already separates the new classes reasonably well, may need no new adapter at all. Whether a new adapter is nevertheless useful can depend on something other than capacity: in exemplar-free learners, class prototypes recorded under an adapter that keeps training become stale (Yu et al., 2020), and freezing that adapter—which is what expansion does—keeps them valid.

This paper asks what common expansion triggers detect, and when what they detect warrants expansion. Our contributions are:

1. **A diagnostic.** We show (Proposition 1) that at a package-level label-novelty event the excess loss of an existing module depends on the input only through its old-label logits, so loss-spike triggers and one-step fresh-module utility respond to classifier novelty, not to representational inadequacy. Empirically, every loss-spike and fresh-utility expansion in our class-incremental streams falls at such an event.

2. **A value-of-expansion protocol.** We measure the value of expansion with oracle expansion at the true boundaries and count-matched random timing, against two references—a single adapter that keeps learning with the same memory budget, and first-session adaptation (Panos et al., 2023; Zhou et al., 2024a)—with trigger thresholds calibrated on shift-free nulls rather than tuned on outcomes.

3. **A regime map.** In the exemplar-based regime, expansion hurts in class-incremental streams yet beats first-session adaptation, so the conclusion of a self-expansion ablation depends on its reference point. In the exemplar-free regime the sign flips [EXT2]. Two mechanism tests support stale classifier statistics as the cause: the flip appears within identical training trajectories when only the prototype rule changes, and exemplar-free drift compensation [EXT4]. The interaction replicates in vision (ViT-B/16, CIFAR-100, ImageNet-R) and inside SEMA's official implementation [SEMA].

4. **A label-conditional detector, and why detection is not enough.** LabelSurprise compares each example's loss with the recent loss of its own label and ignores labels without history. It never expands at class arrivals and fires when established labels change, but its expansions help only where expansion helps.

The study was pre-registered with frozen code hashes in four stages, uses the seed as the unit of analysis, and reports every failed hypothesis. We do not propose a method that competes with the state of the art; our learner is deliberately simple so that the expansion decision is the only moving part.

2 Related work

Expansion in continual learning. Architectural approaches add capacity per task (Rusu et al., 2016; Yoon et al., 2018) or on demand in task-free streams (Lee et al., 2020; Ye & Bors, 2023). With pre-trained models, adapter pools are expanded per task (Zhou et al., 2024b) or when a shift is detected: SEMA uses autoencoder-based representation descriptors (Wang et al., 2025), Online-LoRA monitors loss plateaus (Wei et al., 2025), LACE expands on loss spikes relative to a recent-loss baseline (Tathe, 2026), CABLE scores prospective forgetting after a virtual update (Julian et al., 2025), and LW2G learns whether to grow from a hindrance metric (Feng et al., 2024). These signals are validated by boundary precision—LACE reports that all of its expansions fall at domain boundaries—or by ablations that remove expansion while keeping first-session adaptation, as in SEMA. We stylise the signals of these families inside one learner, evaluate a LACE-style rule with its published threshold, and run SEMA's own code with the expansion decision varied.

Reference points for pre-trained-model continual learning. Adapting a pre-trained model on the first session and freezing it is a strong replay-free baseline (Panos et al., 2023; Zhou et al., 2024a), and prototype classifiers on pre-trained features are strong in general (Rebuffi et al., 2017; Mai et al., 2021; McDonnell et al., 2023). Continual fine-tuning of a single model is competitive when representation learning is slowed and the classifier is re-aligned (Zhang et al., 2023). In exemplar-free learning, the features of old classes drift while the model trains; semantic drift compensation estimates this drift from current data (Yu et al., 2020; Gomez-Villa et al., 2024), and SSIAT keeps a single shared adapter with semantic-shift compensation (Tan et al., 2024). We use these as the two references of the value-of-expansion protocol and as an exemplar-free alternative to expansion.

Classifier effects in class-incremental learning. Output-layer bias toward recent classes is well documented (Wu et al., 2019; Hou et al., 2019; Caccia et al., 2022), and the factorisation of class-incremental prediction into task-identity and within-task prediction is established (Kim et al., 2022). We use the corresponding loss identity as a diagnostic of expansion triggers; the identity itself is not new.

Drift and novelty detection in streams. Error-rate drift detectors (Page, 1954; Gama et al., 2004; Bifet & Gavaldà, 2007) and novel-class detection under concept drift (Masud et al., 2011) distinguish concept change from class emergence in classical stream mining, and class-specific monitoring avoids the masking of drift by class-prior changes (Wang et al., 2013). LabelSurprise transfers this label-conditional view to the expansion decision.

3 Setting and learner

Stream. A task-free learner observes batches $(\mathbf{x}_t, \mathbf{y}_t)$, $t = 1, \ldots, T$, of 16 labelled examples. Segment identities are evaluation metadata and are never shown to the learner, except by the oracle reference policy.

Packages. A frozen encoder f (DistilBERT (Sanh et al., 2019); BERT-base (Devlin et al., 2019) as held-out backbone; ViT-B/16 pre-trained on ImageNet-21k (Dosovitskiy et al., 2021) for vision) is shared. A package k consists of LoRA factors (Hu et al., 2022) on query/value projections (rank 8) and a linear head over the global label space. Only the newest package trains; earlier packages are frozen, as in expand-and-freeze adapter pools (Zhou et al., 2024b; Wang et al., 2025). A new package starts from zero-output LoRA and a zero head.

Training. For every incoming batch the newest package takes two AdamW steps on the batch plus, in the exemplar-based regime, 16 items replayed from its own memory (Chaudhry et al., 2019), with cross-entropy over the labels observed so far.

Memory. A single global reservoir (Vitter, 1985) of 512 examples is shared by the whole pool, so expansion never increases the memory budget.

Two regimes of classifier statistics. Predictions use cosine prototypes; class scores are averaged over packages (equivalently, nearest-class-mean on the normalised concatenation of package features, as in EASE (Zhou et al., 2024b)). In the *exemplar-based* regime, prototypes are recomputed from memory under each package at prediction time, so they always match the current features. In the *exemplar-free* regime (no replay, as in APER, EASE and SEMA), each package accumulates class means of features computed when the examples were observed and never recomputes them; a package spawned after a class was observed holds no prototype for it. Prototypes of a package that keeps training therefore go stale, while those of a frozen package stay valid. (Task-based methods compute prototypes once a task has ended; a task-free learner has no task end, so prototypes accumulate online. Drift compensation, Section 5, is the standard remedy for the resulting staleness.) Every exemplar-free run also reports the exemplar-based rule on the identical trajectory, which isolates the refresh factor.

Learner competence. A single package reaches 85% on the BANKING77 recurrence stream versus 65% for training-free frozen-feature prototypes (Table 1). An earlier version of this project used a learner that reached only 16%—below the training-free baseline—and we discarded its accuracy comparisons (Appendix B).

4 What a loss signal measures at a label-novelty event

Let S be the labels that may be predicted, partitioned into labels O already trained *by the current package* and labels N in the current batch whose head rows in that package have never been updated. N contains labels new to the stream, but also labels seen before the package was created: in an expand-and-freeze pool,

a returning label group is new to the package spawned since it was last seen. We call both *package-level label novelty*. For an example with label y in group $G(y) \in \{O, N\}$,

$$\ell(x, y) = \underbrace{-\log \textstyle\sum_{c \in G(y)} p(c \mid x)}_{\text{label-group term}} \underbrace{- \log p\big(y \mid x, G(y)\big)}_{\text{within-group term}}. \tag{1}$$

Proposition 1. *Suppose the head rows of labels in N have never been updated and are zero, so $z_c(x) = 0$ for $c \in N$. For $y \in N$ the loss of an existing package is*

$$\ell(x, y) = \log |N| \; + \; \log \Big(1 + \tfrac{1}{|N|} \textstyle\sum_{o \in O} e^{z_o(x)}\Big),$$

where the first term is the within-group term and the second is the label-group term. A fresh package with a zero head incurs $\log(|N| + |O|)$. *Hence the advantage of the fresh package before any update (reuse loss minus fresh loss) is* $\log\big((|N| + \sum_o e^{z_o(x)})/(|N| + |O|)\big)$, *which is positive whenever the mean of $e^{z_o(x)}$ over old labels exceeds one, and which depends on x only through the old-label logits.*

Proof. With $z_c = 0$ for $c \in N$, $p(y \mid x) = 1/(|N| + \sum_o e^{z_o})$ and $p(y \mid x, N) = 1/|N|$; the identity follows from equation 1. A zero head assigns $1/|S|$ to every label. □

The proposition is elementary, but its consequence for expansion triggers is direct. A loss-spike detector at a class arrival measures how confidently the old head predicts old labels; a fresh-versus-reuse utility comparison before or shortly after one update rewards discarding that confident head. Neither asks whether the representation can separate the new labels—a question that a growing head and a prototype classifier answer without new adapter capacity. The label-group term is 36–52% of the boundary loss in our class-incremental streams even after several steps of learning (Figure 1).

4.1 A label-conditional alternative

If label novelty is to be absorbed by the classifier, expansion should respond to evidence that an *established* label's mapping has changed. Let B_y be the mean of the last 20 batch-mean losses of label y under the current package.

Definition 1 (LabelSurprise)**.** *For a batch,* $\mathrm{LS} = \frac{1}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} \big(\ell(x_i, y_i) - B_{y_i}\big)$, *where $\mathcal{E}$ contains the examples whose label has at least three prior batch-loss records under the current package. LS is undefined (cannot trigger) if $|\mathcal{E}| < 4$.*

Newly introduced labels are excluded by construction, and while a label is being learned its loss falls, so LS is typically negative under class arrivals. Under concept drift, an established label's loss rises above its own history. LS is an error-based drift statistic (Gama et al., 2004) computed per label. Two earlier variants failed during development (Appendix B).

5 Triggers, references and protocol

Trigger statistics. All triggers are computed on the newest package before it trains on the incoming batch. LabelNovel: fraction of batch labels never observed (a trivial reference). LossZ: z-score of the incoming loss against a 20-batch window (loss-spike triggers; Tathe, 2026; Wei et al., 2025). ReprZ: z-score of a label-free diagonal-Mahalanobis feature distance (distribution-shift descriptors; Wang et al., 2025). InterfLogit: increase in protected-memory cross-entropy after a virtual update (prospective forgetting; Julian et al., 2025). FreshUtil: cross-fitted query-loss advantage of a fresh package over updating the current one after one virtual update (counterfactual utility; Appendix D). Two further statistics, InterfProto and ConflictZ, are failed development proposals reported for completeness, and LabelSurprise is defined above. Virtual updates use exact snapshot/restore of parameters, optimiser moments and random states; a test verifies bitwise restoration.

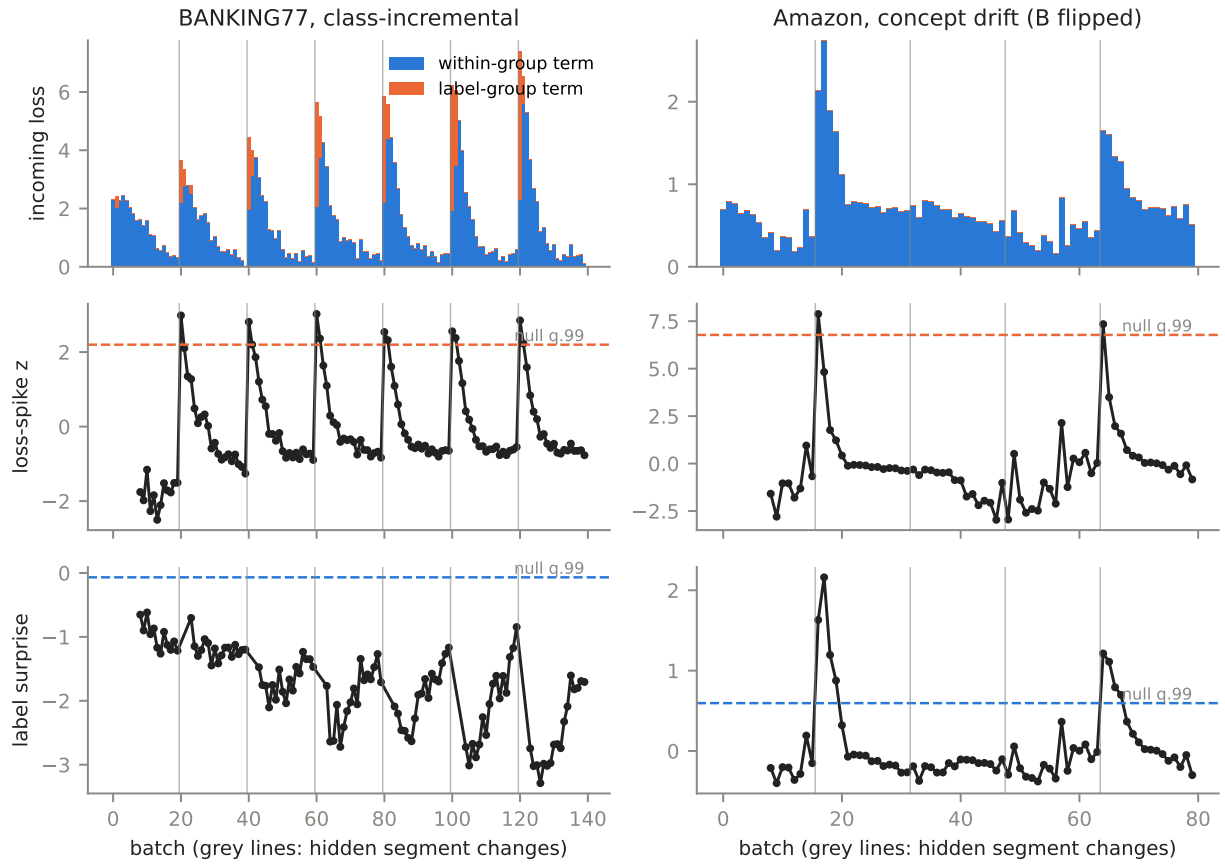


Figure 1: Per-batch statistics of a single package on a class-incremental stream (left) and a concept-drift stream (right); development seed 2026. Top: incoming loss split as in equation 1. Middle: loss-spike z-score. Bottom: LabelSurprise. Dashed: null-calibrated thresholds ($q = 0.99$). The loss spike crosses its threshold at every class arrival; LabelSurprise stays below its threshold under class arrivals and crosses it when established labels flip.

Decision rule and calibration. After the newest package has at least eight updates, a trigger spawns when its statistic exceeds its threshold on two consecutive batches; values in alarm are not added to baseline windows. Each threshold is the 0.99-quantile of the statistic on an i.i.d.-shuffled version of the same stream (same examples, no shift) under the same backbone and regime, using only the development seed. This equalises false-alarm rates across triggers and uses no outcome.

Streams. Class-incremental (CIL): BANKING77 (Casanueva et al., 2020) class-group recurrence (A B A C B) and seven disjoint groups, CLINC150 (Larson et al., 2019) (ten groups of 15), *held-out* 20 Newsgroups (Lang, 1995), Split CIFAR-100 (Krizhevsky, 2009) (ten groups of ten) and ImageNet-R (Hendrycks et al., 2021) (ten groups of 20). Same-label domain shift (DIL): Amazon reviews (Keung et al., 2020) across domains, *held-out* SST-2→Yelp→tweets→IMDB (Socher et al., 2013; Zhang et al., 2015; Barbieri et al., 2020; Maas et al., 2011), and a BREEDS-style (Santurkar et al., 2021) CIFAR-100 superclass stream in which domain d contains the d-th subclass of every superclass. Concept drift: the Amazon and sentiment streams with some domains' labels flipped, the CIFAR superclass stream with two domains relabelled, and a natural label-semantics stream of TweetEval tasks over the same tweet population. Development examples come from training splits only.

Table 1: Final mean development accuracy (%, prototype-mean inference) over ten seeds in the exemplar-based regime; mean number of packages in parentheses. †held-out stream. First-seg.: one package adapted on the first segment, then frozen.

Stream	Frozen	Single	First-seg.	Oracle	Random	LabelNovel	LossZ	ReprZ	InterfLogit	FreshUtil	LS
CIL											
banking_rec	65.0	85.3	69.9	77.1 (5.0)	79.2 (5.0)	81.1 (2.9)	79.5 (4.1)	83.2 (2.8)	83.1 (2.4)	77.3 (5.0)	85.3 (1.0)
banking_cil	46.4	72.7	55.8	61.3 (7.0)	62.8 (7.0)	64.2 (5.1)	62.2 (6.5)	70.5 (1.7)	67.0 (3.3)	62.2 (6.5)	72.7 (1.0)
clinc_cil	58.5	80.8	64.0	70.9 (10.0)	71.4 (10.0)	74.5 (5.2)	71.0 (10.0)	77.8 (2.3)	74.8 (4.9)	71.2 (9.2)	80.8 (1.0)
news_cil†	44.7	56.7	48.7	49.5 (5.0)	50.4 (5.0)	56.4 (1.1)	49.4 (5.0)	55.1 (1.5)	49.8 (4.7)	49.4 (5.0)	56.7 (1.0)
DIL											
amazon_rec	81.9	87.0	84.6	86.4 (5.0)	86.9 (5.0)	87.0 (1.0)	87.0 (1.0)	86.6 (1.5)	86.9 (1.2)	87.0 (1.0)	87.0 (1.0)
amazon_dil	82.0	86.3	85.6	86.6 (5.0)	86.5 (5.0)	86.3 (1.0)	86.3 (1.0)	86.1 (2.5)	86.3 (1.0)	86.3 (1.0)	86.3 (1.0)
senti_dil†	77.1	81.8	80.1	81.8 (4.0)	81.9 (4.0)	81.8 (1.0)	82.1 (1.6)	82.1 (2.8)	81.9 (1.2)	81.8 (1.0)	81.9 (1.5)
Drift											
amazon_conflict	59.0	60.7	60.7	61.5 (5.0)	61.5 (5.0)	60.7 (1.0)	61.0 (3.2)	61.3 (2.2)	61.4 (1.1)	61.8 (3.8)	61.7 (4.0)
amazon_dilconf	57.0	56.3	57.6	58.2 (5.0)	58.2 (5.0)	56.3 (1.0)	58.3 (5.0)	58.0 (3.1)	57.8 (3.0)	58.4 (4.7)	58.3 (5.0)
senti_conf†	55.1	76.3	54.2	58.8 (4.0)	62.2 (4.0)	76.3 (1.0)	61.9 (3.4)	64.4 (2.6)	70.4 (1.9)	63.3 (2.8)	59.6 (4.0)

Policies. *Frozen*: training-free prototypes. *Single*: one package throughout. *First-segment*: one package trained on the first segment only, then frozen, while memory and prototypes keep updating (first-session adaptation). *Oracle*: spawn at every true segment change. *Random*: the oracle's number of spawns at uniformly random steps. *Triggers*: the statistics above.

Exemplar-free refresh. Semantic drift compensation (SDC; Yu et al., 2020) moves each stored prototype μ_c of the training package by the kernel-weighted mean drift of the current batch, $\Delta_c = \sum_i w_{ic}(z_i^{\text{new}} - z_i^{\text{old}}) / \sum_i w_{ic}$ with $w_{ic} = \exp(-\|z_i^{\text{old}} - \mu_c\|^2 / 2\sigma^2)$, on normalised embeddings with the published $\sigma = 0.3$. We apply it after every update, because a task-free stream has no tasks. SDC changes no training computation and is reported as a further prototype rule on the same trajectories.

Statistics and pre-registration. The study was pre-registered in four stages before any confirmation run of each stage (repository files `notes/tmlr_*prereg.md`, with code hashes): the primary text study (seeds 2027–2036; BERT 2027–2031); a vision, baseline-dependence and natural-drift extension (vision seeds 2027–2034); the exemplar-free regime (text 2027–2036, vision 2027–2034); and ImageNet-R plus drift compensation (5–6 seeds). Development used seed 2026 only. The seed is the unit of analysis. We report paired seed-level mean differences with 95% t-intervals and two-sided Wilcoxon signed-rank tests, Holm-corrected within each hypothesis family; deviations are listed in Appendix C.

6 Results

6.1 What the triggers detect

Spawn attribution. In the class-incremental streams without recurrence, every spawn of LossZ and FreshUtil fell within two batches after a batch containing a never-seen label (BANKING-CIL 55/55 and 55/55, CLINC 90/90 and 82/82, 20 Newsgroups 40/40 and 40/40, CIFAR-100 47/47 and 31/31; pooled over seeds). On the recurrence stream BANKING-REC, only 45% and 50% did, so this pre-registered hypothesis failed there. When a label group returns, its labels are new to the package spawned since they were last seen, so Proposition 1 applies; in an analysis declared after the failure, 100% of LossZ and FreshUtil spawns in every class-incremental stream, BANKING-REC included, fell within two batches of package-level label novelty, against a base rate of 15–30% of eligible batches.

A published threshold. The LACE-style rule (loss above 2.5× its recent mean; Tathe, 2026), applied open-loop to single-package traces (five seeds), reproduces the high boundary precision reported for it: in BANKING-CIL, 20 Newsgroups and CIFAR-100 every firing fell within two batches of a segment change (27/27, 20/20, 27/27), each also a batch with a never-seen label, and on BANKING-REC 5/5 firings did. On CLINC, 32 of 52 firings did. In the domain-shift and drift streams it fired rarely (2–12 times for 12–20 changes) and never near label novelty, which those streams lack; there, 2/8 to 12/12 firings were near a

change. High boundary precision in class-incremental streams is therefore what a loss-ratio rule delivers on its own, consistent with Proposition 1; it does not tell whether the resulting expansion pays (Section 6.2).

6.2 The value of expansion with exemplars

Class-incremental: expansion hurts against a learner that keeps training... Oracle expansion at the true boundaries reduced final accuracy relative to a single package by 8.2 points on BANKING-REC, 11.4 on BANKING-CIL, 9.9 on CLINC and 7.1 on held-out 20 Newsgroups (every seed), and 8.9 [7.9, 10.0] on CIFAR-100 with ViT-B/16 (every seed of eight). Count-matched random-timing expansion was also harmful, and on BANKING-REC less so than the oracle (79.2% vs. 77.1%).

...but helps against first-session adaptation. The same oracle pools beat the first-segment policy by 7.2, 5.5 and 6.9 points (every seed) and by 0.9 points on 20 Newsgroups (n.s.), and by 16.6 [13.4, 19.8] on CIFAR-100 (every seed). A self-expansion ablation of the usual form would therefore conclude that expansion helps, in exactly the setting where it costs 7–11 points against a single package that keeps learning, which itself beats first-session adaptation by 15–17 points (Figure 2b).

Held-out backbone. With BERT-base (five seeds; declared secondary), the harm of expansion replicated in every class-incremental stream and every seed: oracle − single was −8.5 [−9.7, −7.3] on BANKING-REC, −9.7 [−11.5, −8.0] on BANKING-CIL, −7.9 [−9.4, −6.4] on CLINC and −5.5 [−6.4, −4.6] on 20 Newsgroups, and LossZ lost 6.2–7.8 points. With five seeds the smallest attainable Wilcoxon p is 0.0625, so no contrast passes the Holm criterion; the t-intervals exclude zero. On the drift streams expansion did not help BERT either (−1.3, −2.1 and −12.7 points), and LabelSurprise again spawned no package in any class-incremental or domain-shift stream.

Without rehearsal. Removing replay from the training loss (memory still feeds the prototypes) reduced the harm of expansion: oracle−single was −4.3 on BANKING-REC (significant) and −3.3, −0.6, −1.2 on BANKING-CIL, CLINC and 20 Newsgroups (n.s.), and +1.4 (n.s.) on CIFAR-100, so our pre-registered prediction that the harm does not depend on rehearsal was mostly not supported. The difference-in-differences (without minus with replay) was +3.9 to +10.3 points in every class-incremental stream (significant, every seed of CIFAR-100).

Domain shift and drift. Under same-label domain shift, oracle expansion was within half a point of the single package. Under concept drift its value was mixed: +1.9 points [0.5, 3.3] on Amazon-DILCONF, +0.8 [−2.1, 3.8] (n.s.) on Amazon-CONFLICT, and −17.5 [−20.5, −14.5] on the held-out sentiment drift stream, the opposite of our pre-registered prediction. In that stream the domains are distinguishable, a single package trained with replay learns the domain-conditional mapping (82% on the flipped Yelp segment at the end), and frozen packages trained on the unflipped mapping out-vote the correct package under prototype averaging (16%). On the natural annotation-drift stream (TweetEval tasks sharing labels; descriptive), oracle expansion was −1.1 points [−2.2, 0.1] and no trigger differed from the single package by more than 0.3 points. On CIFAR-100 domain shift and label-flip drift, oracle expansion lost 2.5 and 13.2 points under prototype averaging (every seed), but *gained* 2.7 [1.7, 3.8] and 4.8 [4.1, 5.5] points under owned-prototype routing (every seed; the drift contrast was added by a dated addendum before any vision confirmation run, after development showed the same reversal). Under drift, whether expansion helps thus depends on how packages are combined at inference, a third axis next to the reference point and the refreshability of classifier statistics.

6.3 The value of expansion without exemplars

With exemplar-free prototypes, the sign of the value of expansion reverses (ten seeds per stream, Holm correction within each family). Oracle expansion beat the single package in all four class-incremental text streams, by 24.0 points [21.3, 26.8] on BANKING-REC, 29.8 [25.2, 34.4] on BANKING-CIL, 28.1 [22.5, 33.7] on CLINC-CIL and 18.4 [12.1, 24.7] on 20 Newsgroups, winning every seed (R1). Within the same runs, the difference between the exemplar-free and the exemplar-based value of expansion was 19.6–33.1 points

(R2, every seed): the training trajectories are identical, so the reversal is caused by the classifier statistics alone. The boundary-detecting LossZ trigger, which lost to the single package in the exemplar-based regime, beat LabelSurprise by 18.0–29.1 points (R3) and matched oracle expansion to within 0.9 points; LabelSurprise, which spawns essentially no package, was indistinguishable from the single package. Oracle placement beat random placement of the same number of packages by 6.1–27.4 points (R4), and the first-segment policy came within 4.5 points of oracle expansion (it beat oracle by 1.6 points on BANKING-CIL and 0.6 on 20 Newsgroups, n.s.). Under same-label domain shift every policy stayed within 1.7 points of the single package; under concept drift, expansion gained 3.4–4.4 points on Amazon-DILCONF and was mixed on Sentiment-CONF. With ViT on CIFAR-100 (eight seeds) the reversal was as large: oracle expansion beat the single package by 31.4 points [27.0, 35.8] (R1), the within-run difference-in-differences was 30.0 [26.0, 34.0] (R2), and LossZ beat LabelSurprise by 31.0 [27.1, 35.0] (R3), every seed. Placement mattered less there: random placement of the same number of packages also gained 27.4 points, and oracle − random (R4) was +3.9 [−4.7, 12.6], not significant. On CIFAR-100 domain shift and drift, exemplar-free expansion gained 0.9 and 2.6 points.

Refreshing without exemplars. If expansion helps exemplar-free learners only because a single package's stored prototypes go stale, an exemplar-free refresh should recover the single package and shrink the value of expansion (five seeds per stream). Drift compensation refreshed the single package in three of five streams (S1: +19.6 points [13.2, 26.0] on BANKING-REC, +10.8 [4.2, 17.3] on 20 Newsgroups, +5.4 [0.7, 10.0] on BANKING-CIL) but *lowered* it on CLINC-CIL (−8.6 [−17.0, −0.1]) and CIFAR-CIL (−19.8 [−22.2, −17.4]), the two streams flagged as at risk in development. It shrank the value of expansion only where it refreshed well (S2 supported on BANKING-REC, −18.6 points, and 20 Newsgroups, −8.9); on CIFAR-CIL the value of expansion grew by 20.7 points. Whatever the rule, oracle expansion still beat the single package in every stream and every seed: by 7.4–49.4 points under SDC ($\sigma = 0.3$), 11.0–51.0 under $\sigma = 0.2$, and 9.5–49.8 under a uniform mean shift. Published exemplar-free refresh therefore repairs part of the staleness on text streams with small drift but does not remove the value of freezing; the first-segment policy remained within 7.4 points of oracle expansion under every rule (Appendix A).

6.4 A canonical benchmark and a published system

ImageNet-R. On the standard pre-trained-model benchmark (200 classes in ten groups of 20; six seeds), both regimes reproduce. With exemplar-refreshed prototypes, oracle expansion lost 18.3 points [16.2, 20.3] to the single package (I1) yet beat first-session adaptation by 5.5 [4.5, 6.6] (I2), every seed. Without exemplars the sign reversed: oracle expansion beat the single package by 8.1 points [3.8, 12.4] (I3), and within the same runs the value of expansion was 16.4 points [15.2, 17.7] higher under exemplar-free than under exemplar-refreshed prototypes (I4). Drift compensation did not rescue the single package here either (oracle − single +22.1 points under SDC). As on CIFAR-CIL, the exemplar-free first-segment policy was the best policy, beating oracle expansion by 9.6 points [6.8, 12.3]: once statistics cannot be refreshed, freezing early beats adapting further.

SEMA. We ran SEMA's official implementation with its published CIFAR-100 configuration and varied only the expansion decision: the official rule; never expanding (adapters frozen after the first task, the reference of SEMA's own ablation); never expanding while the adapters keep training on every task; and expanding at every task. [SEMA results: linear head / exemplar-free NCM vs exemplar NCM.]

6.5 Closed-loop triggers and LabelSurprise

Triggers that detect class boundaries spawned 3–5 packages per class-incremental stream and lost 4–10 points relative to the single package in the exemplar-based regime. LabelSurprise spawned essentially no package in any class-incremental or domain-shift stream, and so matched the single package (non-inferior in all seven), while on the drift streams it spawned 4–5 packages and tracked the oracle: +2.1 [0.9, 3.3] on Amazon-DILCONF, +1.0 (n.s.) on Amazon-CONFLICT and −16.7 on the held-out sentiment drift stream, where expansion itself is harmful. In the exemplar-free regime the ranking reversed: LossZ beat LabelSurprise by 18.0–29.1 points in every class-incremental text stream and seed, and matched oracle

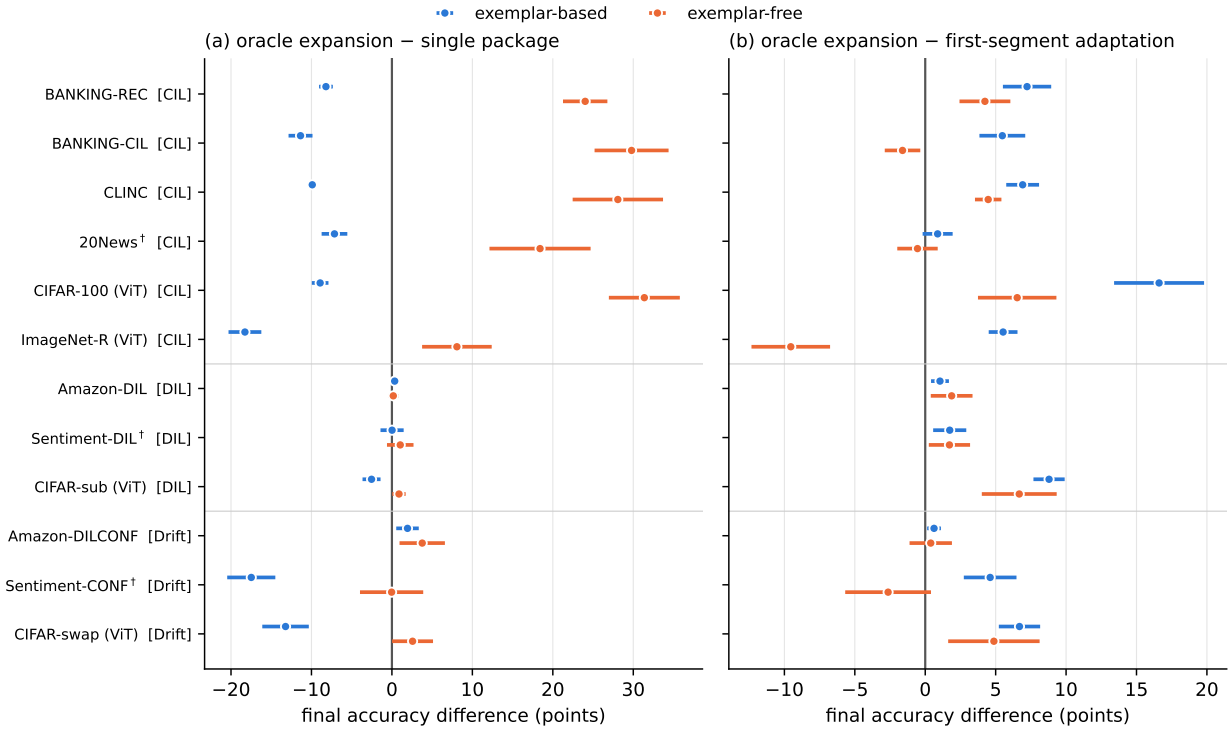


Figure 2: Regime map. Paired seed-level difference in final accuracy (mean, 95% t-interval) between oracle expansion and (a) a single package that keeps learning, (b) first-segment adaptation, with exemplar-refreshed prototypes (blue) and exemplar-free prototypes (orange). Ten seeds for text streams, eight for CIFAR-100, six for ImageNet-R; prototype averaging throughout. †held-out stream.

expansion to within 0.9 points. The pattern held with BERT-base (LossZ −6.2 to −7.8 points against the single package with exemplars; LabelSurprise spawned nothing in class-incremental and domain-shift streams) and with ViT on CIFAR-100: the four boundary-detecting triggers lost 3.0–5.5 points in the class-incremental stream, where LabelSurprise spawned nothing, and on the label-flip drift stream LabelSurprise spawned and tracked the oracle (−12.1 vs. −13.2 under prototype averaging). The one failure of specificity was CIFAR-100 domain shift, where LabelSurprise spawned and lost 2.4 points [1.4, 3.4], failing the pre-registered non-inferiority margin.

7 Discussion and limitations

What expansion buys. [Synthesis: expansion at classifier novelty freezes the features under which classifier statistics were recorded; it pays when those statistics cannot be refreshed and costs when they can. First-session adaptation, drift compensation and exemplar refresh are competing remedies.]

Scope. Our encoders are small to medium, our streams are short (80–500 batches), and three of five drift streams use synthetic relabelling. Our triggers are stylised versions of published signals inside a common learner; the SEMA experiment checks one complete system, and we make no claim about reported benchmark numbers. The value of expansion also depends on the inference rule: routing each input to the package whose own prototypes fit best can recover part of the benefit of expansion under domain shift in vision (Appendix tables report all rules).

Calibration. Thresholds are set on an i.i.d.-shuffled version of the same stream. This equalises false-alarm rates across triggers, which a comparison of triggers requires, but it is not available to a deployed learner.

The conclusion that loss-type triggers fire at label arrivals does not depend on the threshold, since their AUROC for class arrivals is near one.

What LabelSurprise detects. LabelSurprise responds to an increase in the loss of established labels. Concept drift produces this, but so does a within-label shift of $P(x \mid y)$, such as a new subpopulation under an existing superclass. It is therefore a detector of *established-label degradation*, not of concept drift per se.

Training budget of the references. Our first-segment policy trains for the same number of steps per batch as every other policy; SEMA and APER adapt for full epochs on the first task. The SEMA experiment uses SEMA's own budget.

Recommendations. When an expansion trigger is proposed, its evaluation should include (i) the measured value of expansion in the evaluated regime against a single module that keeps learning with the same memory budget, and against first-session adaptation; (ii) a statement of whether classifier statistics are refreshed, since this decides the sign of the value of expansion at class arrivals; (iii) separation of label novelty from changes in established labels, since boundary precision on class-incremental benchmarks is achieved by label-novelty detection alone; and (iv) thresholds calibrated on shift-free nulls rather than tuned on outcomes.

8 Conclusion

[TO BE WRITTEN AFTER CONFIRMATION.]

Use of AI assistants. An AI coding and writing assistant (Claude, Anthropic) was used extensively: to implement experiment code, run and monitor experiments, compute the pre-registered analyses, search and verify related work, and draft and edit text. The author set the research direction, reviewed the code, pre-registration documents, analyses and text, and takes full responsibility for all content. All pre-registrations, code hashes, logs and raw results, including failed and invalid runs, are released with the paper.

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A Extension 4: exemplar-free refresh

Paired differences in final accuracy (points; mean and 95% t-interval over five seeds, 2027–2031), replay 0, exemplar-free prototypes. The refresh rules are inference rules on the same training trajectories.

Stream	Refresh	oracle − single	first-seg − single	oracle − first-seg
BANKING-REC	none	+26.0 [21.4, 30.6]	+21.3 [16.2, 26.3]	+4.8 [2.3, 7.2]
	SDC σ=0.3	+7.4 [1.9, 12.9]	+2.4 [-1.1, 5.9]	+5.0 [2.8, 7.3]
	SDC σ=0.2	+11.0 [5.6, 16.4]	–	–
	mean shift	+10.3 [7.4, 13.3]	–	–
BANKING-CIL	none	+30.2 [18.5, 41.9]	+31.8 [19.9, 43.8]	-1.6 [-4.2, 0.9]
	SDC σ=0.3	+30.4 [20.5, 40.2]	+29.4 [18.9, 40.0]	+0.9 [-1.2, 3.1]
	SDC σ=0.2	+38.5 [30.5, 46.5]	–	–
	mean shift	+26.8 [17.3, 36.4]	–	–
CLINC-CIL	none	+27.8 [19.2, 36.5]	+23.6 [14.7, 32.4]	+4.3 [2.6, 5.9]
	SDC σ=0.3	+36.5 [33.9, 39.2]	+33.3 [31.5, 35.2]	+3.2 [2.2, 4.2]
	SDC σ=0.2	+48.5 [45.4, 51.7]	–	–
	mean shift	+28.0 [25.9, 30.0]	–	–
20NG-CIL	none	+18.8 [8.2, 29.5]	+19.3 [7.7, 30.9]	-0.5 [-3.0, 2.1]
	SDC σ=0.3	+10.0 [2.8, 17.2]	+10.7 [1.1, 20.2]	-0.7 [-3.7, 2.2]
	SDC σ=0.2	+16.9 [6.5, 27.3]	–	–
	mean shift	+9.5 [2.1, 16.9]	–	–
CIFAR-CIL	none	+28.6 [22.7, 34.6]	+21.8 [13.8, 29.7]	+6.8 [1.5, 12.2]
	SDC σ=0.3	+49.4 [45.1, 53.6]	+42.0 [34.7, 49.3]	+7.4 [1.0, 13.8]
	SDC σ=0.2	+51.0 [46.2, 55.9]	–	–
	mean shift	+49.8 [46.4, 53.2]	–	–

B Development history and failed candidates

This project began as the design of a task-free allocator (Appendix D). Its learner took a single AdamW step per 16-example batch into a 77-way head and reached about 16% final accuracy on the BANKING recurrence stream with label-free routing (22% with a learned router), whereas training-free frozen-feature prototypes reached 71% (logistic probe on frozen features: 86%). We therefore discarded all of its accuracy comparisons and built the learner of Section 3. All development of the new testbed used seed 2026 only:

1. InterfProto (head-equalised prospective interference) was our first proposal. On the seed-2026 shadow traces it did not detect concept drift (AUROC 0.54 and 0.22 on the two development drift streams).

2. ConflictZ v1 restricted the loss z-score to examples with previously observed labels. It detected class-incremental boundaries (AUROC 1.00) because labels observed one batch earlier are still untrained. v2 required labels to be first observed at least eight batches earlier; it still fired in class-incremental streams, because newly learned labels have higher loss than long-established ones. This motivated the label-conditional baseline of LabelSurprise.

3. The first closed-loop pilot added alarmed values to the z-score baseline windows, so a genuine spike inflated its own baseline. Excluding alarmed values applies identically to all z-type triggers.

4. A slower single learner (learning rate 10^{-4}) did not reduce exemplar-free staleness in development (BANKING-CIL: 13.1% vs. 26.2% at 10^{-3}) and was not carried into the confirmation.

The thresholds, learner hyperparameters (learning rate 10^{-3}, two steps per batch, replay 16, memory 512) and all trigger constants were fixed before the confirmation study.

C Pre-registration deviations

All pre-registration files, code hashes and results logs are in the repository. Deviations and post-hoc analyses:

- *Primary study.* The package-level novelty analysis of Section 6.1 was declared after hypothesis H1 failed on BANKING-REC and is exploratory. The $q = 0.95$ sensitivity grid (declared secondary) was not run for compute. The BERT grid (declared secondary) was run as a strict subset: single, oracle, LossZ and LabelSurprise, all streams, seeds 2027–2031. Its first execution was invalid and discarded after its outputs were found at chance: the re-created model cache lacked BERT's tokenizer files, and the library silently built an empty vocabulary (accuracy at chance everywhere). After restoring the files at the pinned revision, the development run reproduced bit-for-bit and the subset was re-run unchanged. Twenty-four runs that failed with GPU out-of-memory errors were re-run unchanged; runs are deterministic (a saved run reproduces bit-for-bit).

- *Vision extension.* An addendum declared before any vision confirmation run added the owned-prototype rule for the vision drift stream (development had shown oracle below single under prototype averaging but above it under owned prototypes).

- *Exemplar-free extension.* Thresholds were re-calibrated on rehearsal-free null traces, as declared. A duplicate copy of its pipeline ran for about two minutes; three outputs were written by the duplicate (identical by determinism).

- *ImageNet-R and drift compensation.* Five to six seeds instead of ten, and no triggers on ImageNet-R, for compute (declared). Development observations that SDC lowered the single package's accuracy on CIFAR-100 were recorded in a dated addendum after the freeze, without changes.

- *SEMA.* Per-epoch test evaluation disabled, TF32 matrix multiplication, and a no-gradient forward in expansion detection, identical across variants; the expand-at-every-task variant was dropped before launch.

D Case study: an exact counterfactual allocator

Before this study, we developed and audited an allocator, CAU (counterfactual action utility), that compares an exact, optimiser-faithful one-step update of each existing package and of a fresh classifier-bearing package against a common no-update reference on cross-fitted query halves, subject to a separate protected-memory feasibility check, and spawns after two consecutive windows of positive fresh utility. On a BANKING77 class-group recurrence stream (A B A C B), CAU spawned exactly at the arrivals of the B and C label groups in all ten seed/order development trajectories. A *head-only* variant—identical, except that LoRA outputs are held at zero and only the classifier trains—made identical allocation decisions in all ten trajectories. At the original spawn windows, about 87% of the fresh-versus-best-reuse utility advantage was already present

before the prospective update, i.e. it reflected the replacement of the old classifier rather than learning. These observations, made with the weaker learner of Appendix B and reported only for decisions, motivated Proposition 1 and the present study.